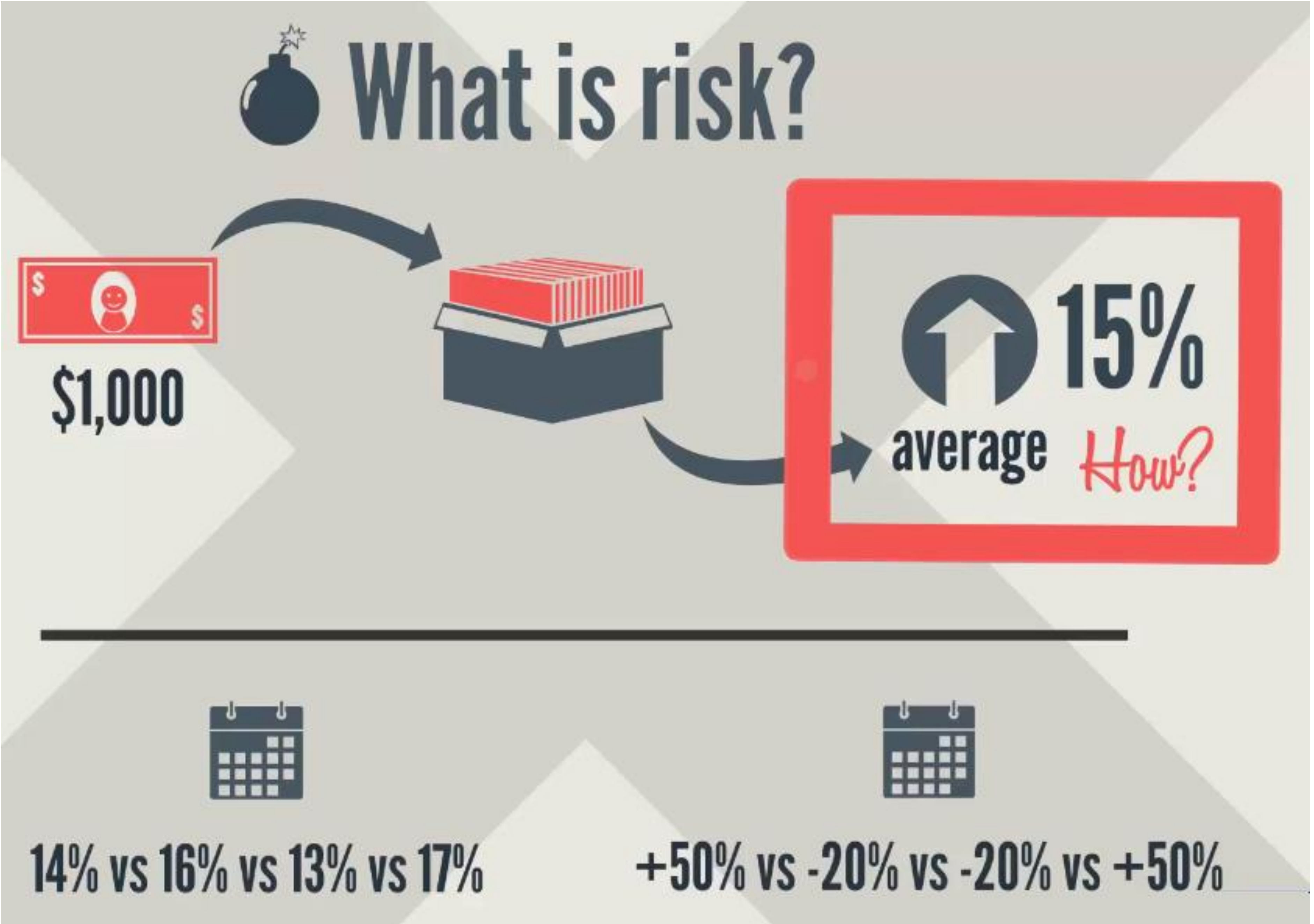
Bài 6 – Phân tích và Quản lý đầu tư nâng cao

Đo lường sự rủi ro

(Measuring Investment Risk)

How do we measure a security's risk?

Security's risk – How to measure and forecast

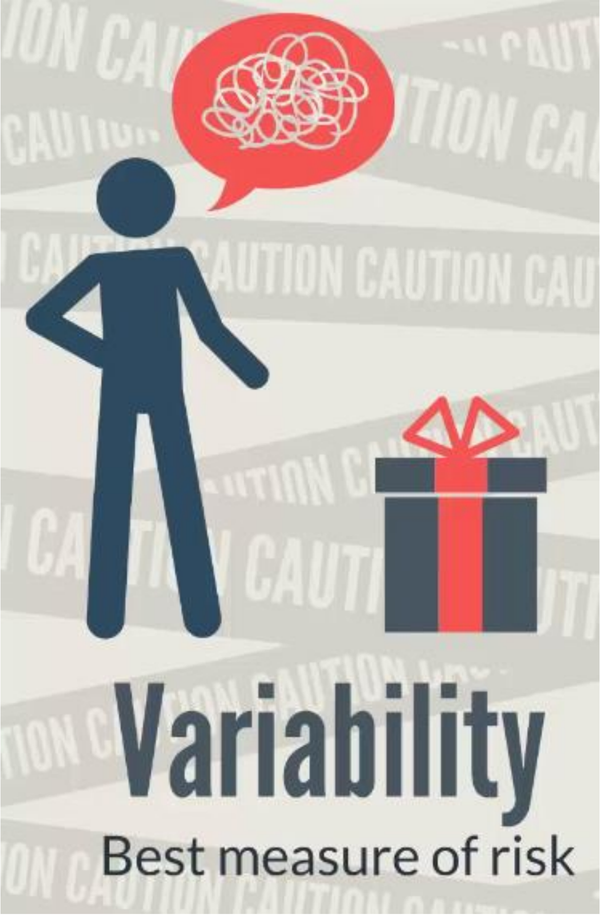


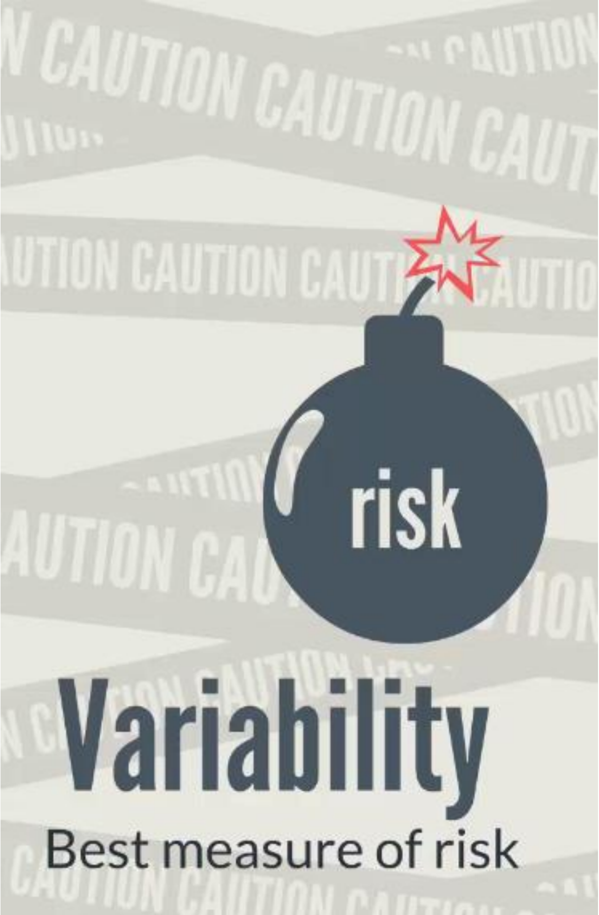
What is risk?


Your money will earn a stable amount over time
14% vs 16% vs 13% vs 17%
Large variability from year to year
Stand to lose 40% of the stock value
+50% vs -20% vs -20% vs +50%

What is risk?


known rate
of return
Variability
Best measure of risk


Variability
Best measure of risk


risk
Variability
Best measure of risk

Statistical measures – to quantify risk (1)

$$s^2 = \frac{\sum (X - \overline{X})^2}{N-1}$$

S^2 Variance

S Standard deviation

measures the dispersion of a set of data points around the mean

$\bar{x} = 15\%$	$(14\% - 15\%)^2$	$= 0.01\%$		$s^2 = 0.1\%/(4-1)$
$x = 14\%$ $x = 16\%$ $x = 13\%$ $x = 17\%$	$(16\% - 15\%)^2$	$= 0.01\%$	$\sum = 0.1\%$	$s^2 = 0.033\ \%$
	$(13\% - 15\%)^2$	$= 0.04\%$		$s^2 = 0.00033$
	$(17\% - 15\%)^2$	$= 0.04\%$		

$$\sqrt{s^2} = \sqrt{0.00033} = 1.8\%$$

Statistical measures – to quantify risk (2)

$$s^2 = \frac{\sum (X - \overline{X})^2}{N-1}$$

S^2 Variance

S Standard deviation

measures the dispersion of a set of data points around the mean

$\bar{x} = 15\%$

$x = +50\%$
$x = -20\%$
$x = -20\%$
$x = +50\%$

$(50\% - 15\%)^2 = 12\%$
$(-20\% - 15\%)^2 = 12\%$
$(-20\% - 15\%)^2 = 12\%$
$(50\% - 15\%)^2 = 12\%$

$\sum = 0.48\%$

$s^2 = 48\%/(4-1)$
$s^2 = 16\ \%$
$s^2 = 0.16$

$\sqrt{s^2} = \sqrt{0.16} = 40\%$

A lot riskier!

Quiz

Which of the following sentences is true?

- The variance equals the square root of the standard deviation
- The standard deviation equals the square root of the variance (correct)
- The variance is the only meaningful statistical measure for measuring risk
- Stock’s return and stock’s risk are two interchangeable financial term

Pratice

Lab05_01_Calculating a Security’s Risk

```
[6] import numpy as np
    import pandas as pd
    from pandas_datareader import data as wb
    import matplotlib.pyplot as plt
    import yfinance as yf

    yf.pdr_override()
```

```
[7] tickers = ['PG', 'BEI.DE']

    sec_data = pd.DataFrame()

    for t in tickers:
        sec_data[t] = wb.DataReader(t, start='2007-1-1')['Adj Close']
```

```
[*********************100%%**********************]  1 of 1 completed
[*********************100%%**********************]  1 of 1 completed
```

```
sec_data.tail()
```

Date	PG	BEI.DE
2023-12-26	145.940002	NaN
2023-12-27	146.059998	134.600006
2023-12-28	145.729996	135.000000
2023-12-29	146.539993	135.699997
2024-01-02	148.740005	135.000000

```
sec_returns = np.log(sec_data / sec_data.shift(1))
```

```
sec_returns
```

Date	PG	BEI.DE
2007-01-03	NaN	NaN
2007-01-04	-0.007621	0.006545
2007-01-05	-0.008624	-0.020772
2007-01-08	0.002202	0.000202
2007-01-09	-0.002517	-0.022858
...	...	...
2023-12-26	0.004533	NaN
2023-12-27	0.000822	NaN
2023-12-28	-0.002262	0.002967
2023-12-29	0.005543	0.005172
2024-01-02	0.014901	-0.005172

4279 rows × 2 columns

PG

```
sec_returns['PG'].mean()
```

```
0.0003094746241432026
```

```
[12] sec_returns['PG'].mean() * 250
```

```
0.07736865603580065
```

```
[13] sec_returns['PG'].std()
```

```
0.011729154703635178
```

```
[14] sec_returns['PG'].std() * 250 ** 0.5
```

```
0.18545421945982196
```

Beiersdorf

```
[15] sec_returns['BEI.DE'].mean()
```

```
0.0002676520082350614
```

```
[16] sec_returns['BEI.DE'].mean() * 250
```

```
0.06691300205876535
```

```
[17] sec_returns['BEI.DE'].std()
```

```
0.013568665222286405
```

```
[18] sec_returns['BEI.DE'].std() * 250 ** 0.5
```

```
0.2145394345536996
```

Final Results:

```
[19] print (sec_returns['PG'].mean() * 250)
     print (sec_returns['BEI.DE'].mean() * 250)
```

```
0.07736865603580065
0.06691300205876535
```

```
[21] sec_returns[['PG', 'BEI.DE']].mean() * 250
```

```
PG        0.077369
BEI.DE    0.066913
dtype: float64
```

```
[ ] sec_returns[['PG', 'BEI.DE']].mean() * 250
```

```
PG        0.064619
BEI.DE    0.063664
dtype: float64
```

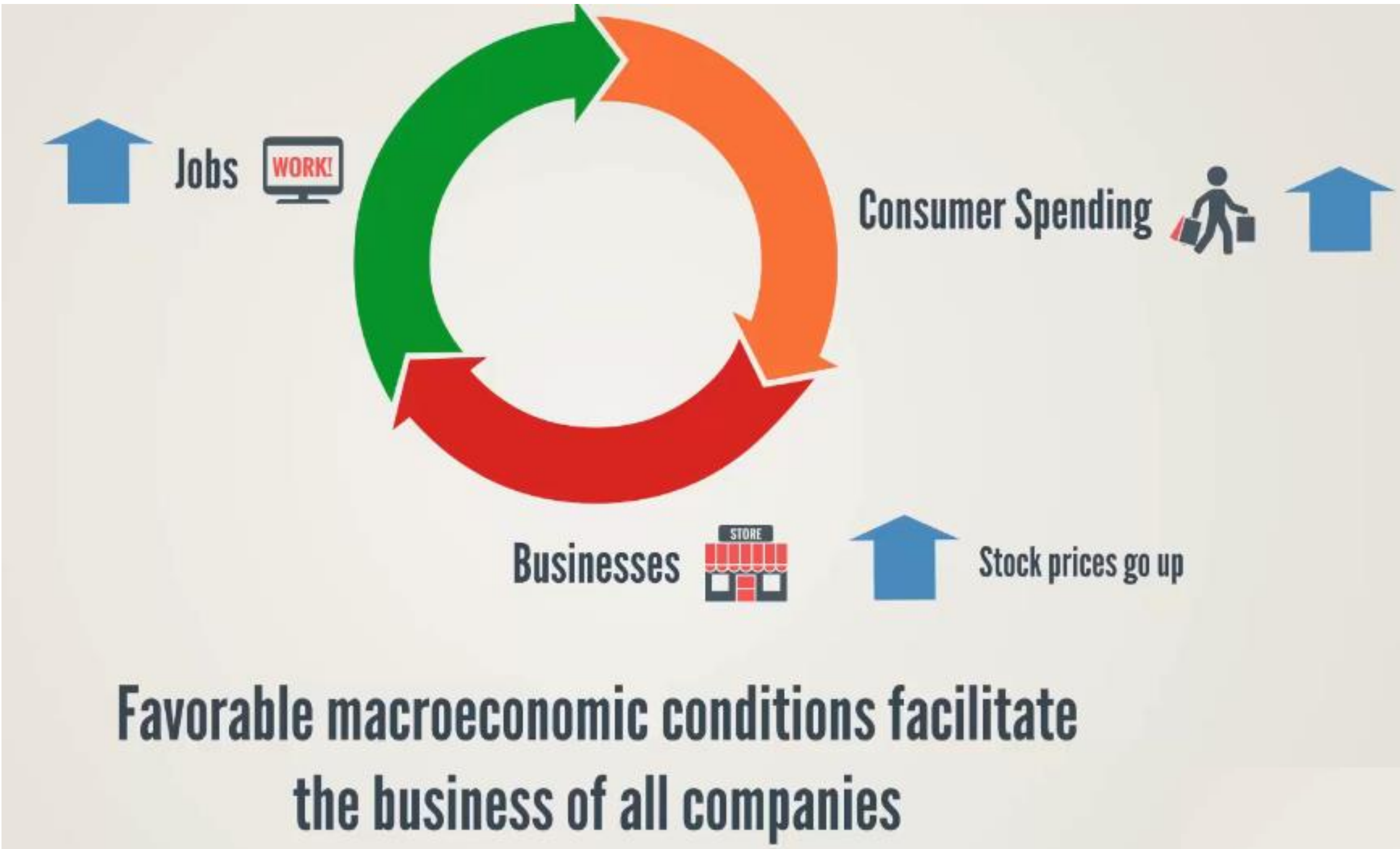
```
[ ] sec_returns[['PG', 'BEI.DE']].std() * 250 ** 0.5
```

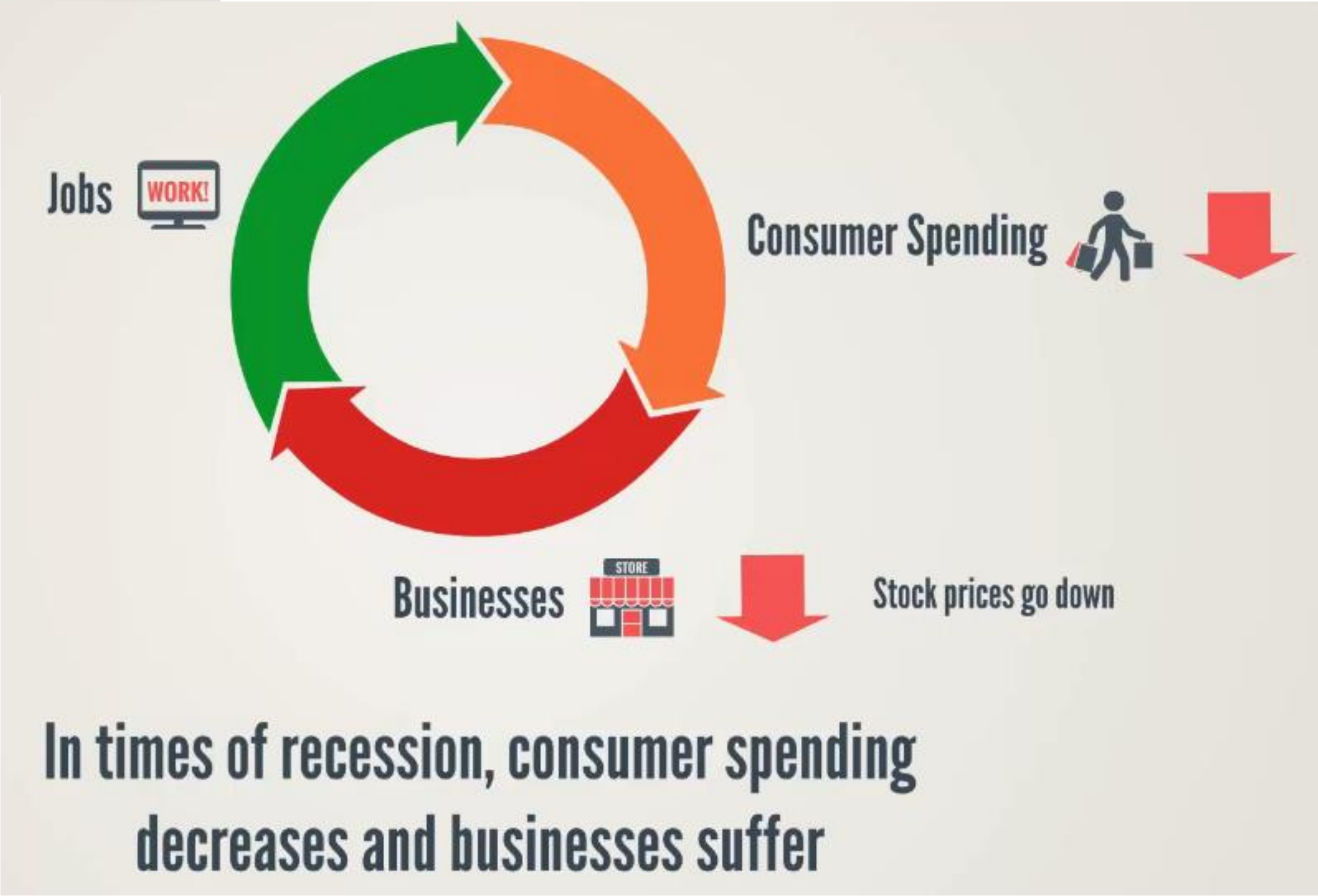
```
PG        0.174245
BEI.DE    0.216454
dtype: float64
```

The benefits of portfolio diversification

Relationship between securities

Macroeconomic Conditions





State of the economic impacts

Share prices are influenced by the state of the economy

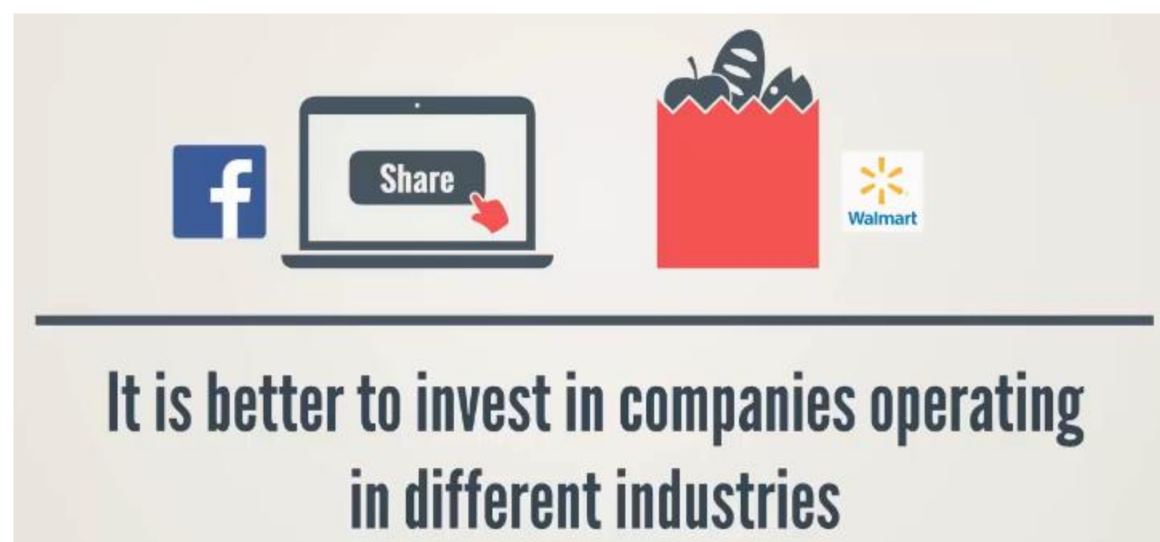
...but different industries are influenced in a different way

It is easier to postpone buying a new car. A weak economy has a stronger impact on the car industry

State of the economic impacts

Relationship between the prices of companies

There is a relationship between the prices of companies,
which will help us optimize investment portfolios



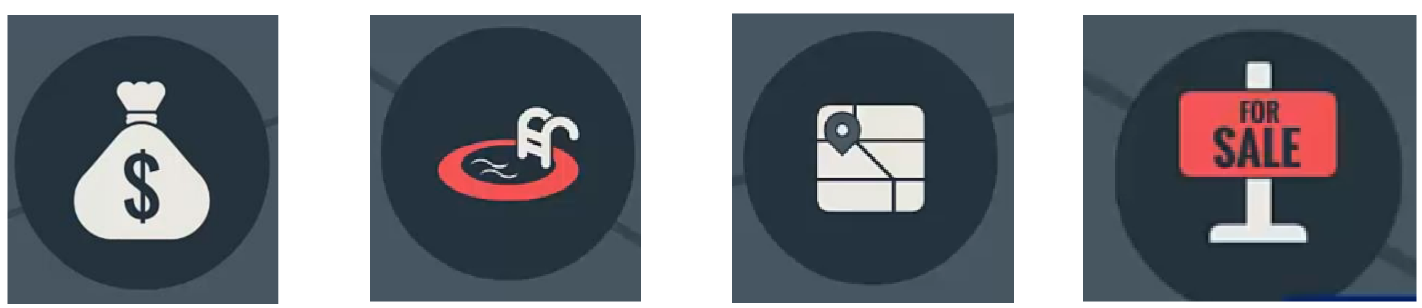
Quiz

Investing in which two stocks will not exploit fully the benefits of portfolio diversification?

- Facebook and LinkedIn (Correct)
- Facebook and Walmart
- LinkedIn and Walmart
- None of the above

Pratice

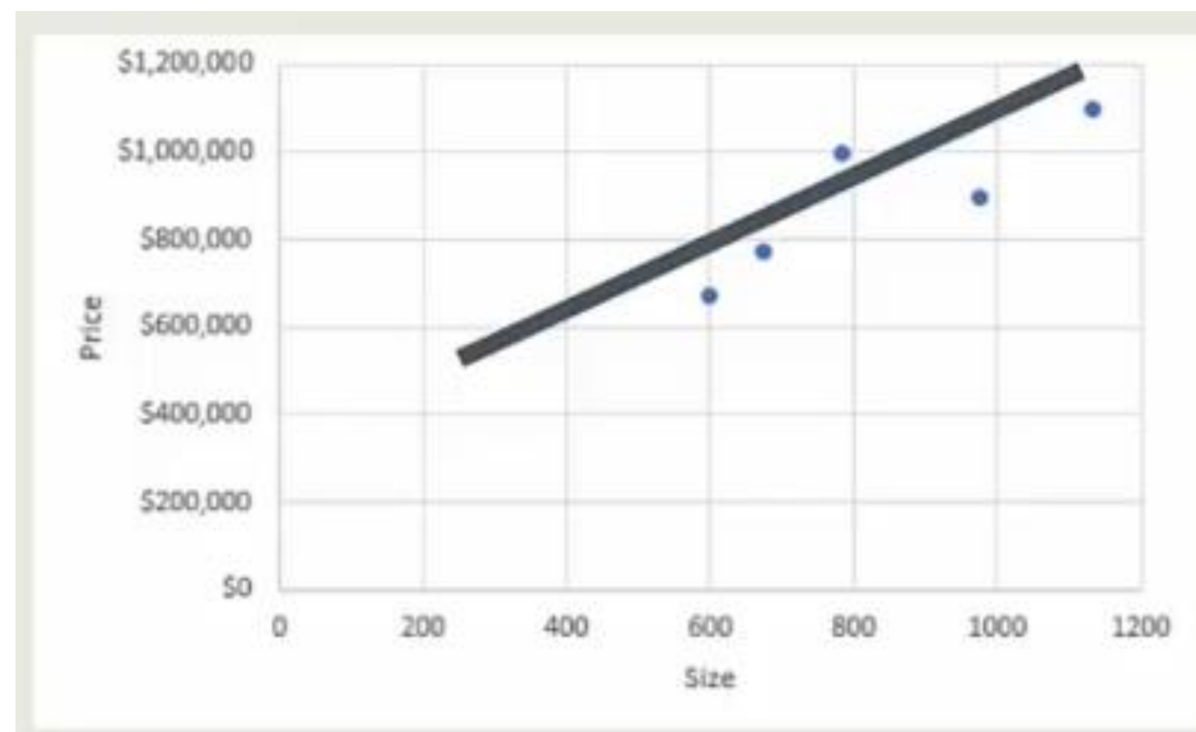
Calculating the covariance



House size – one of the key variables determining house prices

House size – one of the key variables determining house prices

House Listings	
673 sqft	$772,000
785 sqft	$998,000
1,334 sqft	$999,000
699 sqft	$769,000
975 sqft	$895,000



Calculating the covariance

$$\rho_{xy} = \frac{(x - \bar{x}) * (y - \bar{y})}{\sigma_x \sigma_y}$$

x - house size

y - house price

The correlation coefficient measures the relationship between two variables

	Size (X)	Price (Y)		(x-x̄)*(y-ȳ)
	673	$ 772,000		25,234,920
	785	$ 998,000		(12,053,480)
	1334	$ 999,000		49,545,920
	699	$ 769,000		22,837,920
	975	$ 895,000		687,120
Average:	893.2	886600	Sum:	86,252,400
			Cov:	21,563,100

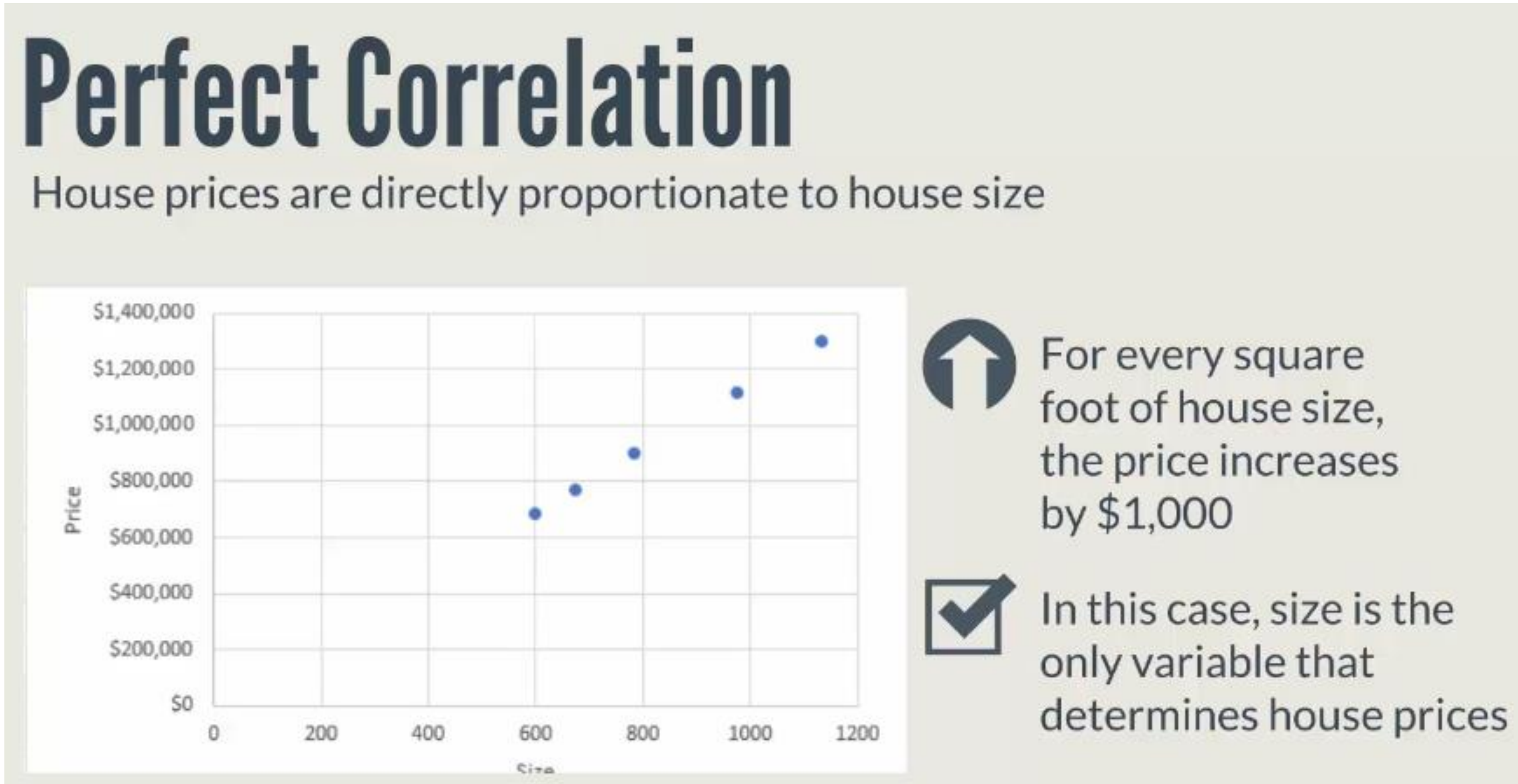
If Covar > 0 --> The two variables move in the same direction

If Covar < 0 --> The two variables move in opposite directions

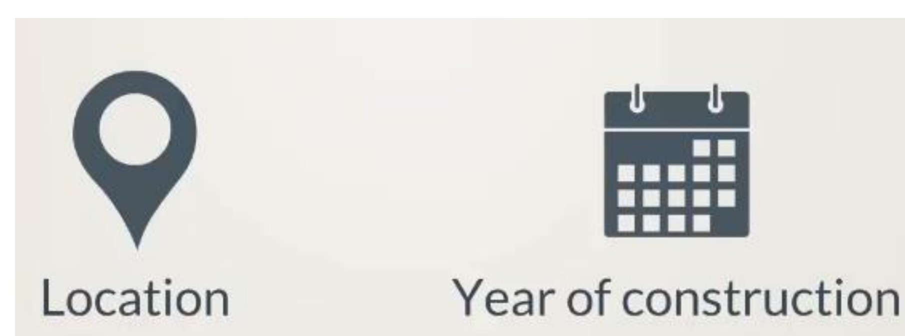
If Covar = 0 --> The two variables are independent

Measuring the correlation between stocks

Interpreting Correlation in House Price

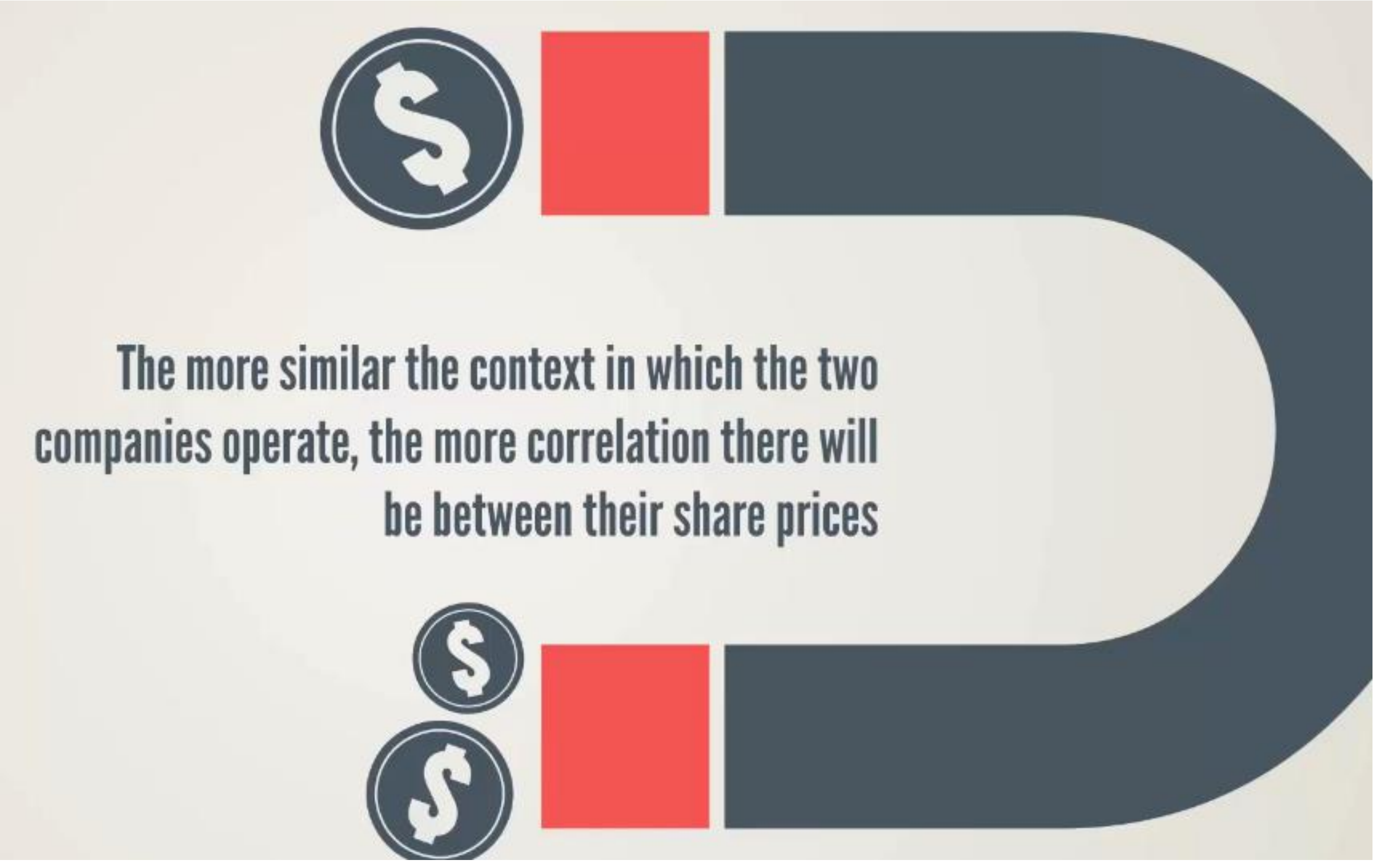


In reality, several variables have an impact on house prices

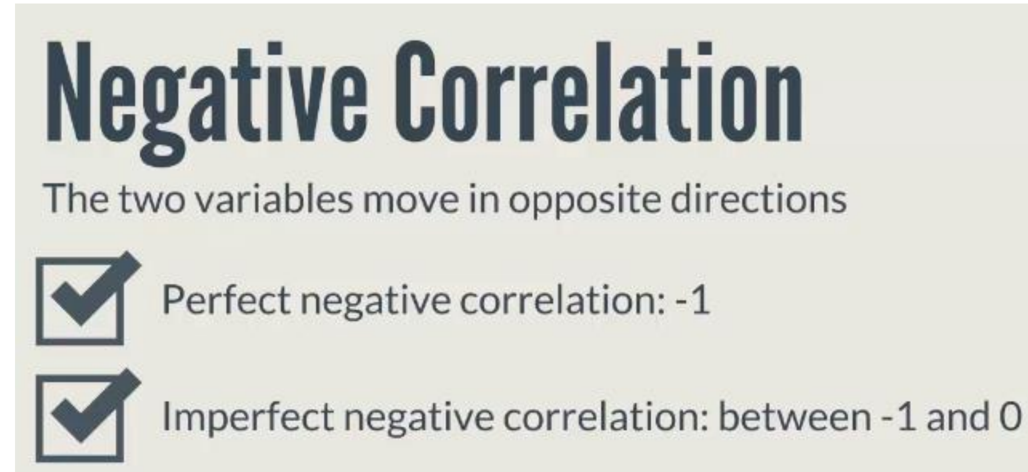


Interpreting Correlation in Share Prices

In the same way, several variables determine share prices:



Types of Correlation



Quiz

Which of the following situations is supposed to exemplify the behavior of two stocks with a negative covariance?

- Their prices are moving in the same positive direction
- Their prices are moving in the same negative direction
- Their prices are moving in the opposite direction (Correct)
- The two companies have gone bankrupt

Quiz

How should a correlation of 1 be interpreted?

- Perfect positive correlation (Correct)
- Independence
- Partial correlation
- Perfect negative correlation

Pratice

Lab05_02_Calculating Covariance and Correlation

Covariance and Correlation

$$CovarianceMatrix:\ \Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1I} \\ \sigma_{21} & \sigma_2^2 & \dots & \sigma_{2I} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{I1} & \sigma_{I2} & \dots & \sigma_I^2 \end{bmatrix}$$

[20]
```
PG_var = sec_returns['PG'].var()
PG_var
```

```
0.00013757322336096703
```

[21]
```
BEI_var = sec_returns['BEI.DE'].var()
BEI_var
```

```
0.00018410860740094673
```

[22]
```
PG_var_a = sec_returns['PG'].var() * 250
PG_var_a
```

```
0.034393305840241 76
```

[23]
```
BEI_var_a = sec_returns['BEI.DE'].var() * 250
BEI_var_a
```

```
0.04602715185023668
```

[24]
```
cov_matrix = sec_returns.cov()
cov_matrix
```

	PG	BEI.DE
PG	0.000138	0.000043
BEI.DE	0.000043	0.000184

[25]
```
cov_matrix_a = sec_returns.cov() * 250
cov_matrix_a
```

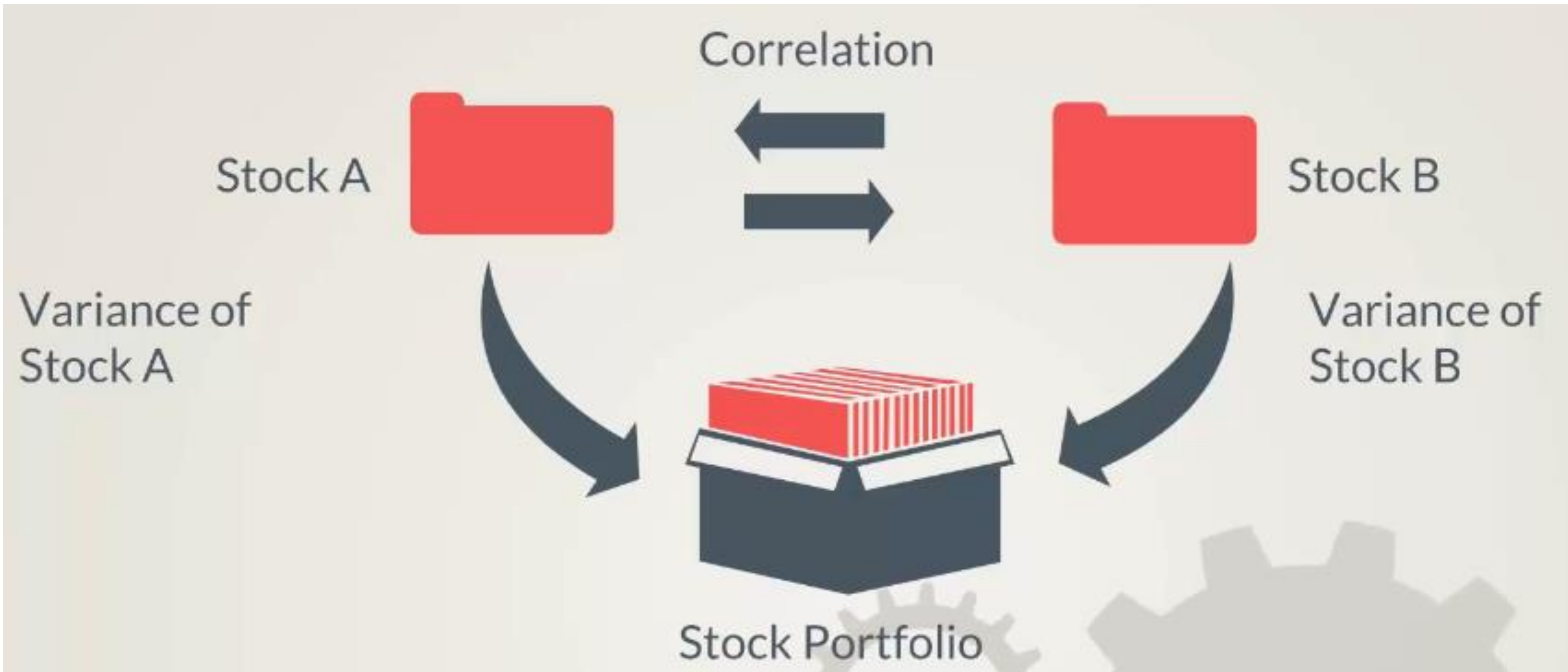
	PG	BEI.DE
PG	0.034393	0.010695
BEI.DE	0.010695	0.046027

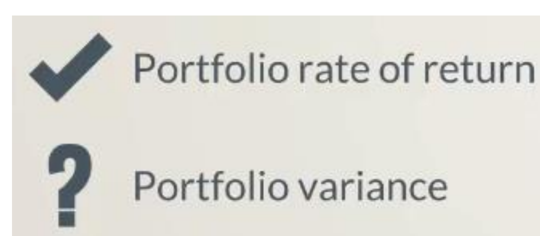
[26]
```
corr_matrix = sec_returns.corr()
corr_matrix
```

	PG	BEI.DE
PG	1.000000	0.268171
BEI.DE	0.268171	1.000000

Considering the risk of multiple securities in a portfolio

Portfolio variance – Calculating the variability of a portfolio of two or more stocks





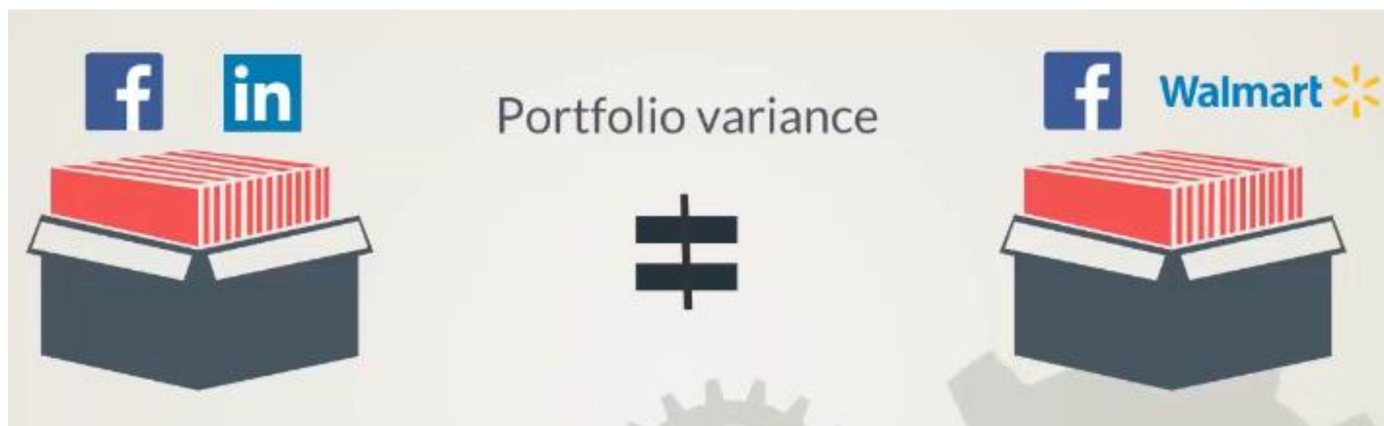
Portfolio variance

Portfolio variance (2 stocks):

$$(w_1\sigma_1 + w_2\sigma_2)^2 = ?$$

$$w_1 + w_2 = 1$$

$$(a+b)^2 = a^2 + 2ab + b^2$$



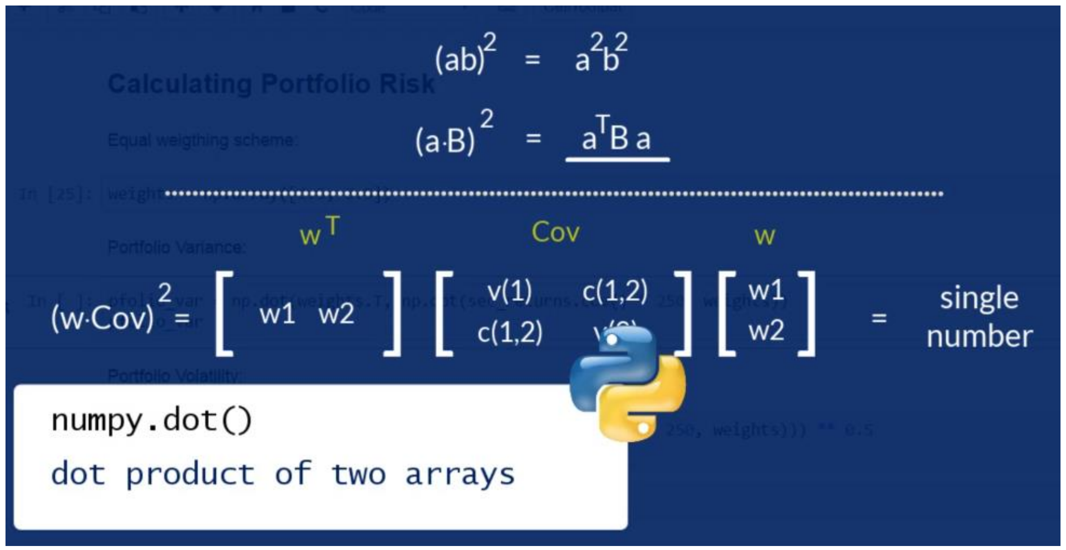
$$(w_1\sigma_1 + w_2\sigma_2)^2 = w_1^2\sigma_1^2 + 2w_1\sigma_1 w_2\sigma_2\rho_{12} + w_2^2\sigma_2^2$$

Main inputs:

1. σ_1, σ_2
2. ρ_{xy}

Pratice

Lab05_03_Calculating Portfolio Risk



Calculating Portfolio Risk

Equal weigthing scheme:

```
[29] weights = np.array([0.5, 0.5])
     weights
```

```
array([0.5, 0.5])
```

Portfolio Variance:

```
[26] pfolio_var = np.dot(weights.T, np.dot(sec_returns.cov() * 250, weights))
     pfolio_var
```

```
0.025452447409699028
```

Portfolio Volatility:

```
[27] pfolio_vol = (np.dot(weights.T, np.dot(sec_returns.cov() * 250, weights))) ** 0.5
     pfolio_vol
```

```
0.1595382318119987
```

```
[28] print (str(round(pfolio_vol, 5) * 100) + ' %')
```

```
15.953999999999999 %
```

Understanding Systematic vs. Idiosyncratic risk

Portfolio Variance

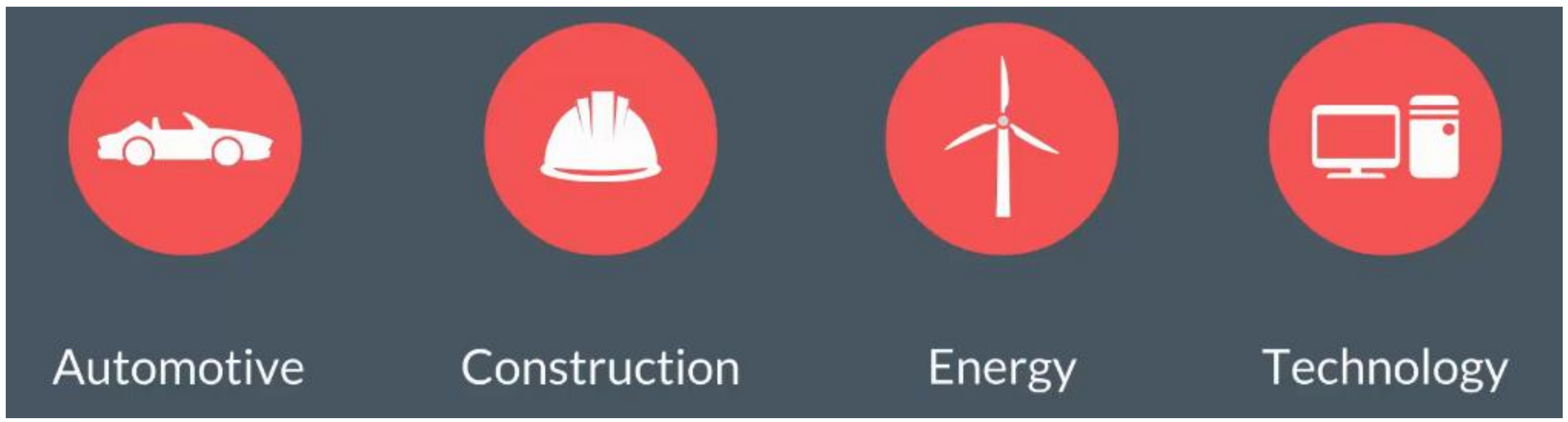
Un-diversifiable risk	**Diversifiable risk**
This component depends on the variance of each individual security. It is also known as systematic risk	Idiosyncratic risk (also known as company specific risk). Driven by company-specific events

Un-diversifiable Risk

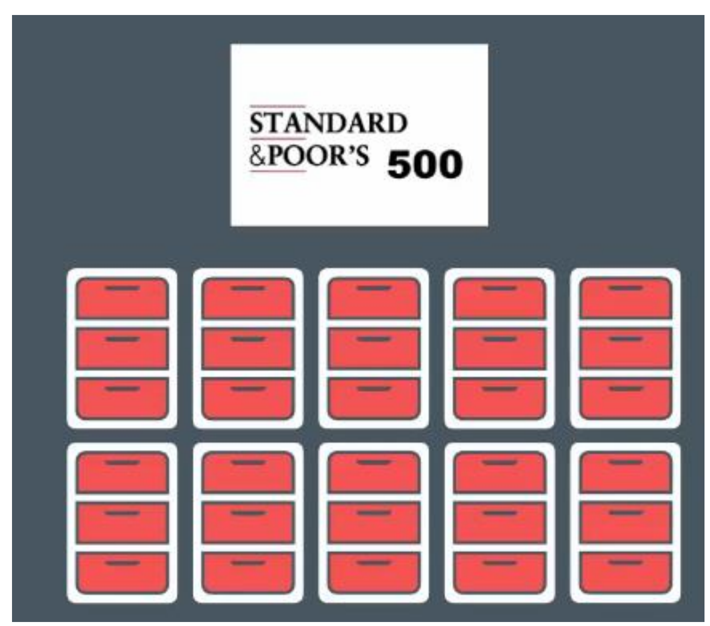
There are a number of systematic risks that cannot be diversified

Diversifiable Risk

Diversifiable risk can be eliminated if we invest in non-correlated assets



Investment vehicles tracking indices like the S&P 500 are well diversified



Quiz

Which of the following is false?

- Undiversifiable risk is a synonym for systematic risk
- Diversifiable risk is a synonym for unsystematic risk
- Undiversifiable risk is a synonym for unsystematic risk (Correct)
- Undiversifiable risk cannot be eliminated

Pratice

Lab05_04_Calculating Diversifiable and Non-Diversifiable Risk of a Portfolio

Calculating Diversifiable and Non-Diversifiable Risk of a Portfolio

```
[30] weights = np.array([0.5, 0.5])
```

```
[31] weights[0]
```

```
0.5
```

```
[32] weights[1]
```

```
0.5
```

Diversifiable Risk:

```
[33] PG_var_a = sec_returns[['PG']].var() * 250
     PG_var_a
```

```
PG     0.034393
dtype: float64
```

```
[34] BEI_var_a = sec_returns[['BEI.DE']].var() * 250
     BEI_var_a
```

```
BEI.DE    0.046027
dtype: float64
```

```
[45] dr = pfolio_var - (weights[0] ** 2 * PG_var_a) \
          - (weights[1] ** 2 * BEI_var_a)
     dr
```

```
0.005347326153478995
```

```
float(PG_var_a)
```

```
0.03439331483717425
```

```
[37] PG_var_a = sec_returns['PG'].var() * 250
     PG_var_a
```

```
0.03439331483717425
```

```
[38] BEI_var_a = sec_returns['BEI.DE'].var() * 250
     BEI_var_a
```

```
0.0460271662433006
```

```
[44] dr = pfolio_var - (weights[0] ** 2 * PG_var_a) \
         - (weights[1] ** 2 * BEI_var_a)
     dr
```

```
0.005347326153478995
```

```
[40] print (str(round(dr*100, 3)) + ' %')
```

```
0.535 %
```

Non-Diversifiable Risk:

```
[41] n_dr_1 = pfolio_var - dr
     n_dr_1
```

```
0.020105120270118 71
```

```
[42] n_dr_2 = (weights[0] ** 2 * PG_var_a) + (weights[1] ** 2 * BEI_var_a)
     n_dr_2
```

```
0.020105120270118713
```

```
[43] n_dr_1 == n_dr_2
```

```
False
```

THANKS FOR YOUR ATTENTION!